

Dylan Thompson

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EDUCATION

University of Central Florida
B.S. Mechanical Engineering

GPA 4.0
Expected Graduation: 2029

EXPERIENCE

GE Vernova Supply Chain Manufacturing Intern

May 2026– Aug 2026

- Collaborating with cross-functional teams to optimize the production of high-voltage grid components, focusing on improving assembly efficiency and reducing material waste.
- Scheduled to apply Lean Manufacturing and Six Sigma principles to streamline floor operations and enhance First Pass Yield (FPY) for critical grid infrastructure.

Undergraduate Researcher – Interfacial Transport Lab

Jun 2025 – Present

- Investigating experimental methods to improve the efficiency and predictability of pulsating heat pipes (PHPs), focusing on single-loop aluminum prototypes that use two-phase fluid oscillations for passive thermal transport.
- Conducted independent CFD analysis of a single-loop aluminum pulsating heat pipe (PHP) using ANSYS software to evaluate fluid flow characteristics and thermal performance for research optimization.
- Developed a SolidWorks exploded-view model of the aluminum PHP, collaborated with UCF machinists on prototype fabrication, and co-authored the research paper abstract.
- Debugged Python scripts for improved oscillation data visualization and thermal performance analysis.

American Society of Mechanical Engineers (ASME) Project Lead, 2025–26 Student Design Competition

Sept 2025 – Present

- Leading a multidisciplinary team (25+ members) in designing and fabricating an autonomous waste collection robot for ASME's 2025-2026 Student Design Competition.
- Coordinating CAD modeling, prototyping, and physical testing to optimize device navigation, waste sorting accuracy, and material handling efficiency across varied terrain types including inclines and rough surfaces.
- Managing comprehensive design validation and deliverables, including fabrication documentation and demonstration videos, to maximize scoring potential and ensure rule compliance during the timed Initial and Elimination Testing Demonstrations.

Baja SAE Machinist

Aug 2025 – Present

- Machined precision components for Baja SAE competition vehicles using CNC lathe (ProtoTRAK), manual mill, and fabrication equipment to meet rigorous performance specifications and tight tolerances.
- Prototyped, inspected, and documented parts to ensure reliability and compliance with SAE competition rules, contributing to successful builds for endurance, acceleration, and maneuverability events.

Robotics Club of Central Florida (Project Storm) | Drive & Science Team Designer |

Aug 2025 – Dec 2025

- Designed and tested mechanical systems for a Mars Rover prototype intended for planetary desert navigation and autonomous object retrieval using a robotic arm.
- Fabricated custom wooden test rigs using precision bandsaw cuts and accurately drilled holes to securely fit screws, enabling motor load testing up to 125 lbs. and validation of structural integrity and power delivery.
- Utilized precision CAD modeling to improve servo integration and ensure reliable operation across articulated subsystems.

PERSONAL PROJECTS

Robotic Arm with Arduino Control

Jul - Aug 2025

- Designed and fabricated a 4-axis robotic arm in SolidWorks, fully modeled with custom modifications.
- Integrated Arduino + servo driver board to control multiple servos.
- Implemented potentiometer-based manual control for real-time positioning.

SKILLS

- **Certifications:** Lean Six Sigma Yellow Belt, Certified SolidWorks Professional (CSWP), MOS Certified (Excel, PowerPoint, Word)
- **Programming & Electronics:** Python, Arduino (C++), Servo Systems, Feedback Control Software: ANSYS (CFD), ProtoTRAK (CNC).
- **Interpersonal:** Communication, Leadership, Problem-Solving, Project Management, Adaptability.

ACTIVITIES AND HONORS

- Co-Founder, asxw Clothing (Designed and launched brand generating \$100k+ in sales).
- DECA Senator and Competitor (Leadership & Business Skills).
- UCF Skateboarding Club